

D701

Elementary Mathematics Methods and Interventions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

Items

1.

A fourth grader in a classroom with a new document camera is asked to mark $\frac{2}{3}$ on a number line labeled 0, 1, and 2. She divides the whole line from 0 to 2 into three equal parts and marks the second tick. The class's fraction unit began Monday. What is the error?

- A. The task cannot be done without fraction strips.
- B. She should have divided the line into two parts.
- C. She partitioned the whole line instead of the distance from 0 to 1, so her mark is at $\frac{4}{3}$.
- D. She should have counted tick marks from 1.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
-----------------------------	---	----------------------	--

2.

A fifth grader who is saving for a bike says, " $12 \times \frac{1}{2}$ can't be 6, because multiplying always makes a number bigger." He has just earned a perfect score on multiplication facts. What is the best teacher response?

- A. Explain that multiplying by fractions always makes numbers smaller.
- B. Tell him the rule only holds for whole numbers and move on.
- C. Use a model to show $12 \times \frac{1}{2}$ as half of 12 groups and discuss why a factor less than 1 makes the product smaller.
- D. Agree, since his facts are perfect.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
-----------------------------	---	----------------------	--

3. [TRANSFER — reversed stem / different variable]

What is $6 \div \frac{1}{2}$, and why?

- A. 6, because dividing by a fraction changes nothing.
- B. $\frac{1}{12}$, because you flip the answer.
- C. 12, because there are 12 halves in 6.
- D. 3, because division makes numbers smaller.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
-----------------------------	---	----------------------	--

4.

In a sixth-grade class, which has a large window facing the playground, a student says, “I think 0.4 is bigger than 0.35 because tenths are bigger than hundredths.” The lesson is being observed by the principal. What teacher response best builds discourse?

- A. Pick a quieter topic since the principal is watching.
- B. “Correct, great job,” then move to the next problem.
- C. “Let me explain why that works.”
- D. “Who can put that reasoning in their own words, and do you agree or disagree?”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A third grader who always turns in homework on time writes $52 - 17 = 45$, $63 - 28 = 45$, and $71 - 34 = 43$. She uses a sparkly pencil. What is the most likely misconception?

- A. She regroupes but forgets to subtract the tens.
- B. She subtracts the smaller digit from the larger in each column instead of regrouping.
- C. She is adding instead of subtracting.
- D. She is making careless errors.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

If the same student solves $61 - 28$, what answer does her misconception predict, and what is correct?

- A. She will write 43; the correct answer is 33.
- B. She will write 33; the correct answer is 33.
- C. She will write 89; the correct answer is 33.
- D. She will write 47; the correct answer is 33.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A fifth-grade teacher with 28 students, including several who are well above and several well below grade level, wants one lesson on area. The school recently adopted a new curriculum. Which task best fits?

- A. Wait to teach area until the new curriculum's pacing guide arrives.
- B. A worksheet of 20 length-times-width problems for everyone.
- C. Easier computation sheets for low students and harder ones for high students.
- D. "Find as many rectangles as you can with an area of 24 square units. Which has the smallest perimeter?"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A fourth grader who plays chess after school finally solves a multi-step problem after trying two different drawings. He wears his favorite team jersey that day. Which feedback best supports a growth mindset?

- A. "When your first drawing didn't work, you tried another one. That persistence paid off."
- B. "Good job."
- C. "Chess players are always good at math."
- D. "You're a natural at math."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student gives up on a hard problem and says, "I'm just not a math person." What is the best response?

- A. "That's okay, not everyone is a math person."
- B. "You're not there yet. Let's try drawing it or starting with a smaller case."
- C. "Skip it; hard problems are never worth the stress."
- D. "You're smart, so you can do it."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A Tier 2 group of four sixth graders is still unable to solve ratio problems after a unit of open-ended exploration. The group meets in a converted storage room with good lighting. What change is best supported by research?

- A.** Add explicit modeling with think-alouds and tape diagrams, guided practice with feedback, then independent practice.
- B.** Assign more independent worksheets.
- C.** Move the group to a bigger room.
- D.** Give more open exploration time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A third grader in a class that loves building with pattern blocks says a tilted square “is a diamond, not a square, because it looks like a diamond.” The class has geometry on Wednesdays. At what Van Hiele level is the student reasoning?

- A.** Visualization (Level 0)
- B.** Formal deduction, since the student gave a reason.
- C.** Informal deduction (Level 2)
- D.** Analysis (Level 1)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A teacher wants to move students from visualization to analysis. Which activity fits?

- A.** Write two-column proofs about triangles.
- B.** Sort shapes by properties, such as “has a right angle” or “has two pairs of parallel sides,” and explain each sort.
- C.** Name shapes on flashcards as fast as possible.
- D.** Show every shape only in its standard position.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A fourth grader ~~in a classroom with a new document camera~~ is asked to mark $\frac{2}{3}$ on a number line labeled 0, 1, and 2. She divides the whole line from 0 to 2 into three equal parts and marks the second tick. ~~The class's fraction unit began Monday.~~ What is the error?

Target word: name the error **Decides:** The unit is 0 to 1, not 0 to 2.

A	Absolute language	A number line is a valid length model on its own.
B	Wrong stem number	Uses the numerator as the number of parts.
C	Correct	Thirds must split the unit interval; two of her parts reach $\frac{4}{3}$.
D	Inverted operation	Counting starts at 0, and spaces are counted, not ticks.

2. Answer C Concept 2

A fifth grader ~~who is saving for a bike~~ says, " $12 \times \frac{1}{2}$ can't be 6, because multiplying always makes a number bigger." ~~He has just earned a perfect score on multiplication facts.~~ What is the best teacher response?

Target word: address the misconception **Decides:** Multiplying by less than 1 gives a smaller product.

A	Absolute language	Multiplying by a fraction greater than 1 (like $\frac{3}{2}$) makes it larger.
B	Dropped element	Drops the model that builds understanding.
C	Correct	Addresses the overgeneralized rule with meaning.
D	Camouflage	Fact mastery does not make the claim true.

3. Answer C Concept 2 · transfer

What is $6 \div \frac{1}{2}$, and why?

Target word: division by a fraction less than 1 **Decides:** How many halves are in 6?

A	Dropped element	Ignores the divisor.
B	Confusion set	Flips the wrong thing; you multiply by the reciprocal of the divisor.
C	Correct	Dividing by a number less than 1 gives a larger quotient.
D	Inverted operation	Divides by 2 instead of by $\frac{1}{2}$.

4. Answer D Concept 3

In a sixth-grade class, ~~which has a large window facing the playground,~~ a student says, "I think 0.4 is bigger than 0.35 because tenths are bigger than hundredths." ~~The lesson is being observed by the principal.~~ What teacher response best builds discourse?

Target word: best talk move **Decides:** Ask others to engage with the reasoning.

A	Camouflage	The observer does not decide the best move.
B	Dropped element	Drops the discussion of reasoning.
C	Inverted operation	The teacher takes over the thinking.
D	Correct	Restate and agree/disagree moves get others reasoning.

5. Answer B Concept 4

A third grader ~~who always turns in homework on time~~ writes $52 - 17 = 45$, $63 - 28 = 45$, and $71 - 34 = 43$. ~~She uses a sparkly pencil.~~ What is the most likely misconception?

Target word: find the pattern **Decides:** In each ones place she subtracts the smaller digit from the larger.

A	Inverted operation	She does not regroup at all; the tens were subtracted.
B	Correct	$7 - 2 = 5$ and $5 - 1 = 4$ gives 45; the others follow the same bug.
C	Confusion set	$52 + 17 = 69$, not 45.
D	Absolute language	The errors follow a consistent pattern, so they are not careless.

6. Answer D Concept 4 • transfer

If the same student solves $61 - 28$, what answer does her misconception predict, and what is correct?

Target word: predict the bug's output **Decides:** Ones: $8 - 1 = 7$; tens: $6 - 2 = 4$.

A	Wrong stem number	Copies an earlier answer rather than applying the pattern.
B	Dropped element	Ignores the bug.
C	Inverted operation	89 is the sum, not her pattern.
D	Correct	Smaller-from-larger gives 47; with regrouping, $61 - 28 = 33$.

7. Answer D Concept 5

A fifth-grade teacher with 28 students, including several who are well above and several well below grade level, wants one lesson on area. ~~The school recently adopted a new curriculum.~~ Which task best fits?

Target word: task for a wide range **Decides:** An open task with many entry points and extensions.

A	Camouflage	The adoption does not decide the task.
B	Dropped element	One path with no entry points or extension.
C	Adjacent category	Tracks students into different content instead of one rich task.
D	Correct	Everyone can start; it extends to comparing perimeter.

8. Answer A Concept 6

A fourth grader ~~who plays chess after school~~ finally solves a multi-step problem after trying two different drawings. ~~He wears his favorite team jersey that day.~~ Which feedback best supports a growth mindset?

Target word: best feedback **Decides:** Praise the strategy and persistence.

A	Correct	Names the process and effort that led to success.
B	Dropped element	Drops the specific process.
C	Camouflage	Pulls on the chess detail and praises a fixed trait.
D	Inverted operation	Praises a fixed trait, the opposite of growth-mindset feedback.

9. Answer B Concept 6 · transfer

A student gives up on a hard problem and says, “I’m just not a math person.” What is the best response?

Target word: response to struggle — reverse of praise **Decides:** Reframe ability as growable and give a strategy.

A	Inverted operation	Confirms the fixed-ability belief.
B	Correct	Frames ability as growing and offers a next strategy.
C	Absolute language	Avoids productive struggle.
D	Adjacent category	Trait praise, not strategy support.

10. Answer A Concept 7

A Tier 2 group of four sixth graders is still unable to solve ratio problems after a unit of open-ended exploration. The group meets ~~in a converted storage room with good lighting.~~ What change is best supported by research?

Target word: best intervention change **Decides:** Struggling learners need explicit, systematic instruction.

A	Correct	Explicit instruction with visual models is effective for students with math difficulties.
B	Adjacent category	Practice without modeling or feedback.
C	Camouflage	The room does not decide the instructional change.
D	Dropped element	Repeats the approach that did not work and drops explicit teaching.

11. Answer A Concept 8

A third grader ~~in a class that loves building with pattern blocks~~ says a tilted square “is a diamond, not a square, because it looks like a diamond.” ~~The class has geometry on Wednesdays.~~ At what Van Hiele level is the student reasoning?

Target word: name the level **Decides:** Judging by appearance and orientation.

A	Correct	Identifies shapes by how they look.
B	Confusion set	Giving a reason is not a proof; the reason is appearance.
C	Inverted operation	Would reason about relationships among classes; the student ignores properties.
D	Adjacent category	Analysis uses properties such as equal sides and right angles.

12. Answer B Concept 8 · transfer

A teacher wants to move students from visualization to analysis. Which activity fits?

Target word: activity that lifts the level **Decides:** Analysis means describing shapes by properties.

A	Adjacent category	Formal deduction skips levels students have not reached.
B	Correct	Focuses attention on properties, the analysis level.
C	Dropped element	Keeps students recognizing by look.
D	Inverted operation	Reinforces judging by orientation, keeping students at visualization.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Fractions on the number line	A fraction is a number with one location on the number line: split the distance from 0 to 1 into equal parts named by the denominator and count those parts.	Splitting the whole drawn line (0 to 2) instead of the unit, or counting tick marks instead of spaces.	A number line that goes past 1.
2	Multiplying and dividing by fractions	Multiplying by a number less than 1 makes a smaller product; dividing by a number less than 1 makes a larger quotient.	"Multiplication always makes bigger, division always makes smaller."	A fraction or decimal less than 1 as a factor or divisor.
3	Productive talk moves	Moves like wait time, revoicing, asking others to restate or add on, and pressing for reasoning build discourse where students explain and critique ideas.	Confirming the right answer quickly and moving on.	What the teacher says right after a student shares.
4	Error analysis of student work	Look for a consistent pattern in errors (such as always subtracting the smaller digit from the larger) to find the misconception, then reteach the concept.	Calling a systematic error careless and just assigning more practice.	Several wrong answers from the same student with a shared pattern.
5	Low-floor, high-ceiling tasks	Open tasks that everyone can start and that extend far let all students work on the same big idea at different depths.	Differentiating by giving struggling students only easier computation worksheets.	One task with many entry points or solutions.
6	Growth-mindset feedback	Praise effort, strategy, and progress ("You tried another way when the first did not work") rather than fixed traits ("You are so smart").	Believing praise for intelligence builds confidence and persistence.	What a teacher says after success or struggle.
7	Explicit instruction for math intervention	Students with math difficulties benefit from explicit, systematic instruction: clear modeling with think-alouds, guided practice with feedback, then independent practice, using visual representations.	Relying only on unguided discovery for students who are already behind.	Designing Tier 2 or Tier 3 math support.
8	Van Hiele levels of geometric thinking	Students move from visualization (recognizing by look) to analysis (by properties) to informal deduction (relationships among properties and classes); instruction should match and lift the level.	Assuming students can follow class-inclusion logic when they still identify shapes by appearance.	"It looks like a..." vs. "It has four right angles, so..."

Course scope (public description): Elementary Mathematics Methods and Interventions is an engaging course crafted for educators who are dedicated to strengthening and enriching the mathematics education of elementary students. This course focuses on practical, research-supported strategies for teaching essential mathematical

concepts, including number operations, geometry, measurement, and data analysis. This course includes resources for all elementary grade levels with a significant focus on grades 3 to 6. Participants learn to identify and assess students' mathematical understanding, create differentiated learning experiences, and implement effective interventions for diverse learner needs. Emphasis is placed on constructing a classroom environment that promotes mathematical curiosity, problem-solving, meaningful discourse, and a growth mindset. Students acquire the skills to build authentic mathematics experiences that caters to the varied needs of all students, setting a firm foundation for their future academic success in mathematics.